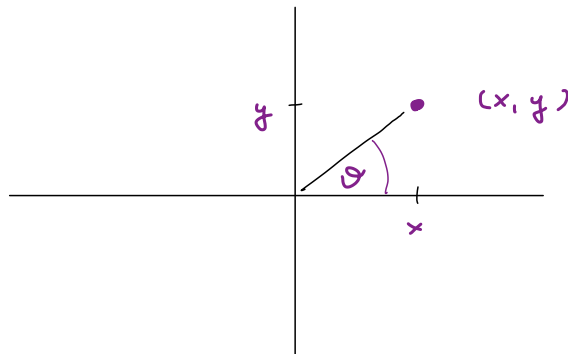


Complex Fourier series and generalized expansions

Complex numbers (revision)

$$x, y \in \mathbb{R} \rightarrow z = (x, y) \in \mathbb{C} \quad \text{complex plane}$$



- addition

$$z_1 + z_2 = (x_1 + x_2, y_1 + y_2)$$

- multiplication

$$z_1 \cdot z_2 = z_2 \cdot z_1 = (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2)$$

commutative, associative, distributive

notation $i \equiv (0, 1) \Rightarrow i^2 = (-1, 0)$

$$z = x + i \cdot y \Leftrightarrow x \cdot (1, 0) + y (0, 1)$$

- inverse

$$-z = -x - i y \quad (\text{addition})$$

$$z^{-1} \equiv \frac{1}{z} = \frac{1}{x + i y} = \frac{x - i y}{(x + i y) \cdot (x - i y)} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$$

- complex conjugation

$$\bar{z} = z^* = x - i y$$

$$z \cdot \bar{z} = |z|^2 = x^2 + y^2$$

- polar representation

$$z = (r \cos \vartheta, r \sin \vartheta) = r \cos(\vartheta) + i r \sin(\vartheta)$$

$$\text{multiplication} \rightarrow r = r_1 \cdot r_2 \quad \vartheta = \vartheta_1 + \vartheta_2 \bmod 2\pi$$

- polar representation

$$z = (r \cos \theta, r \sin \theta) = r \cos(\theta) + i r \sin(\theta)$$

$$\text{multiplication} \rightarrow r = r_1 \cdot r_2 \quad \theta = \theta_1 + \theta_2 \pmod{2\pi}$$

Complex exponential and Euler form

Def: $e^z \equiv e^x (\cos(y) + i \sin(y))$

$$\left(\text{or} \equiv \sum_{n=0}^{\infty} \frac{1}{n!} z^n \right)$$

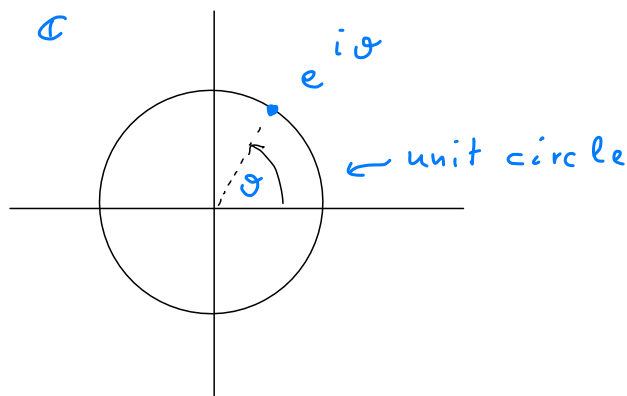
then $e^{i\theta} = \cos \theta + i \sin \theta$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

$$\Rightarrow \cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$z = r \cdot e^{i\theta} = r \cos(\theta) + i \cdot r \sin(\theta)$$

$$z_1 \cdot z_2 = r_1 \cdot r_2 \cdot e^{i(\theta_1 + \theta_2)}$$



usual properties carry over

$$e^{z_1} e^{z_2} = e^{z_1 + z_2}$$

homework: prove it!

Complex Fourier series

learned that if $f(\theta)$ is 2π -periodic then we can approximate it as

$$f(\theta) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n \cdot \theta) + b_n \sin(n \cdot \theta))$$

$$\text{with } a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta f(\theta)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} d\theta f(\theta) \cdot \cos(n \cdot \theta)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} d\theta f(\theta) \cdot \sin(n \cdot \theta)$$

we now rewrite it in complex form

$$f(\theta) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n \cdot \theta) + b_n \sin(n \cdot \theta))$$

$$= a_0 + \sum_{n=1}^{\infty} a_n \frac{1}{2} (e^{in\theta} + e^{-in\theta}) - b_n \frac{1}{2} i (e^{in\theta} - e^{-in\theta})$$

$$= a_0 + \sum_{n=1}^{\infty} \left(\underbrace{a_n}_{c_n} \frac{1-i}{2} e^{in\theta} + \underbrace{\frac{a_n+ib_n}{2}}_{c_{-n} = c_n^*} e^{-in\theta} \right)$$

$$= c_0 + \sum_{n=1}^{\infty} c_n e^{in\theta} + \sum_{n=1}^{\infty} c_{-n} e^{-in\theta}$$

make one big sum

$$\downarrow$$
$$= \sum_{n=-\infty}^{\infty} c_n e^{in\theta}$$

$$\sum_{n'=-\infty}^{-1} c_{n'} e^{in'\theta}$$

To summarize

$$(*) \quad f(\vartheta) = \sum_{n=-\infty}^{\infty} c_n e^{in\vartheta} \quad \Leftarrow \quad \text{complex Fourier series}$$

with

$$c_{n>0} = \frac{a_n - ib_n}{2}; \quad c_{n<0} = \frac{a_n + ib_n}{2} = \overline{c_{n>0}}$$

$$c_{n=0} = a_0$$

The c_n 's can be expressed as a simpler but complex-valued integral ...!

$$\text{Statement:} \quad c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\vartheta \, f(\vartheta) e^{-in\vartheta} \quad (**)$$

Proof 1

$$\begin{aligned} n > 0 \quad c_n &= \frac{1}{2} (a_n - ib_n) = \\ &= \frac{1}{2} \left(\frac{1}{\pi} \int_{-\pi}^{\pi} d\vartheta \, f(\vartheta) \cos(n\vartheta) - i \frac{1}{\pi} \int_{-\pi}^{\pi} d\vartheta \, f(\vartheta) \sin(n\vartheta) \right) \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} d\vartheta \, f(\vartheta) \underbrace{(\cos(n\vartheta) - i \sin(n\vartheta))}_{e^{-in\vartheta}} = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\vartheta \, f(\vartheta) e^{-in\vartheta} \quad \checkmark \end{aligned}$$

$$\begin{aligned} n < 0 \quad c_n &= (c_{-n})^* = \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} d\vartheta \, f(\vartheta) \underbrace{e^{-i(-n)\vartheta}}_{e^{+in\vartheta}} \right)^* \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} d\vartheta \, f^*(\vartheta) \cdot \underbrace{(e^{in\vartheta})^*}_{\substack{\uparrow \\ f(\vartheta) \text{ real}}} = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\vartheta \, f(\vartheta) e^{-in\vartheta} \quad \checkmark \end{aligned}$$

$$n = 0 \quad c_0 = a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\vartheta \, f(\vartheta) \quad \checkmark$$

□

Proof 2 :

step 1 : first prove orthogonality relation :

$$\int_{-\pi}^{\pi} d\theta e^{-im\theta} e^{in\theta} = 2\pi \delta_{n,m}$$

$$e^{z_1} e^{z_2} = e^{z_1+z_2}$$

$$\int_{-\pi}^{\pi} d\theta e^{-im\theta} e^{in\theta} = \int_{-\pi}^{\pi} d\theta e^{i(m-n)\theta} =$$

$$\stackrel{\text{Euler}}{=} \int_{-\pi}^{\pi} (\cos((m-n)\theta) - i \sin((m-n)\theta)) d\theta$$

• for $n=m \Rightarrow \int_{-\pi}^{\pi} d\theta (1 - i \cdot 0) = 2\pi$

• for $k \equiv n-m \neq 0 \rightarrow \cos(k\theta)$ and $\sin(k\theta)$ have $|k|$ periods in $[-\pi, \pi] \Rightarrow$ integrate to 0 ✓

Step 2 :

Assuming $f(\theta) = \sum_{n=-\infty}^{\infty} c_n e^{in\theta}$ I evaluate

the integral $\int d\theta f(\theta) e^{-in\theta}$ and show that

it yields $c_n \cdot 2\pi$.

$$\int d\theta f(\theta) e^{-in\theta} = \int_{-\pi}^{\pi} d\theta \left(\sum_{a=-\infty}^{\infty} c_a e^{ia\theta} \right) e^{-in\theta}$$

use assumption use arbitrary summation label

$$= \sum_{a=-\infty}^{\infty} c_a \underbrace{\int_{-\pi}^{\pi} d\theta e^{ia\theta} e^{-in\theta}}_{\text{yields } 0 \text{ if } a \neq n, 2\pi \text{ if } a=n}$$

exchange $\sum$ and $\int$ $\leftarrow a=n$ term $C_{a=n} \cdot 2\pi = c_n \cdot 2\pi$ $\rightarrow$ only $a=n$ term survives in sum! □

in summary: If (*) holds then c_n
can be expressed as (**)

Remark: • Proof 2 works if $f: \mathbb{R} \rightarrow \mathbb{C}$, but
then $c_{-n} \neq c_n^*$, usually

Extension to other periods

$$f(x + 2L) = f(x)$$

$$\vartheta \leftrightarrow \frac{\pi}{L} \cdot x$$

$$d\vartheta \leftrightarrow \frac{\pi}{L} \cdot dx$$

with this, I get

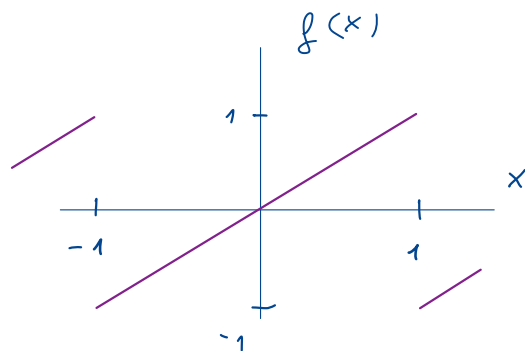
$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i n \frac{\pi}{L} x}$$

$$c_n = \frac{1}{2L} \int_{-L}^L dx f(x) e^{-i n \frac{\pi}{L} \cdot x}$$

Example:

$$f(x) = x \quad x \in [-1, 1]$$

$$f(x+2) = f(x) \quad L = 1$$



$$\Rightarrow c_n = \frac{1}{2} \int_{-1}^1 dx x e^{-i n \cdot \pi \cdot x} = \frac{2}{-}$$

$$I \equiv \int_{-1}^1 dx \cdot x \cdot e^{-i n \cdot \pi \cdot x} = ?$$

$$n=0 \rightarrow \emptyset$$

$$n \neq 0 \quad e^{-i n \cdot \pi \cdot x} = \cos(n \cdot \pi \cdot x) - i \sin(n \cdot \pi \cdot x)$$

$$I = \int_{-1}^1 dx \cdot x \cdot \cos(n \cdot \pi \cdot x) - i \int_{-1}^1 dx \cdot x \cdot \sin(n \cdot \pi \cdot x) =$$

$\uparrow$ odd $\uparrow$ even
 $\Rightarrow \emptyset$

$$= -i \cdot 2 \int_0^1 dx \cdot x \cdot \sin(n \cdot \pi \cdot x) \quad \text{partial int.}$$

$$= (-2i) \left\{ \left[x \left(-\frac{1}{n \cdot \pi} \right) \cos(n \cdot \pi \cdot x) \right]_0^1 + \frac{1}{n \cdot \pi} \int_0^1 dx \cos(n \cdot \pi \cdot x) \right\}$$

$\parallel$
 $-\frac{1}{n \cdot \pi} \cos(n \cdot \pi) = -\frac{1}{n \cdot \pi} (-1)^n$

$$= \frac{2i}{n \cdot \pi} (-1)^n \Rightarrow \underline{C_{n \neq 0} = \frac{i}{n \cdot \pi} (-1)^n}$$

Then

$$f(x) = \sum_{n \neq 0} \frac{i}{n \cdot \pi} (-1)^n e^{i n \cdot \pi \cdot x}$$

remark:

can also rewrite it in terms of sine functions

$$f(x) = 2 \cdot \sum_{n > 0} \frac{(-1)^{n+1}}{n \cdot \pi} \cdot \sin(n \cdot \pi \cdot x) =$$

$$= \frac{2}{\pi} \left(\sin(\pi \cdot x) - \frac{1}{2} \sin(2\pi x) + \dots \right)$$